BUELE

10BASE-T1S-USB-ETH

10BASE-T1S to 10M USB Ethernet Converter

User Manual

Revision History

Date	Version	Description
2024/10/11	V1.0	Initial release

1. Description

1.1 Overview

10BASE-T1S-USB-ETH is a 10BASE-T1S to 10M USB Ethernet converter with metal case, supporting both Multidrop bus line and Point-to-Point topologies. It comes with a DB9 connector (PIN2 → TXN−, PIN7 → TXP+) and is bundled with a DB9-to-Terminal Board featuring PIN2, PIN7 and a 100 Ω termination resistor on/off button, which makes it easy for end customers to use with 10BASE-T1 Ethernet.

10BASE-T1S is a new automotive Ethernet PHY standard following 100BASE-T1 and 1000BASE-T1. With a bandwidth of 10 Mbps, multi-device shared topology, and PLCA (Physical Layer Collision Avoidance) mechanism, 10BASE-T1S provides an Ethernet-based alternative to traditional bus systems. It supports PLCA and conforms to the IEEE 802.3cg standard, using UTP for communication.

Supports a data rate of up to 10 Mbps over a single twisted-pair cable, suitable for up to 25 meters of full/half duplex networks, aiming to achieve collision-free and deterministic transmission on multi-point networks.

1.2 Product Features

- 10BASE-T1 to 10M USB Ethernet
- Speed: 10 Mbps
- Point-to-Point
- Multidrop Support
- PLCA (Physical Layer Collision Avoidance)
- Status LED
- Metal Shell
- DB9 Connector
- USB Power Supply

1.3 Specifications

Parameter	Value
Power Supply	USB power supply; PLCA endpoint control with USB Ethernet adapter
Power Consumption	100 mW
Data Rate	10 Mbps
Operating Temperature	−40°C to +85°C

Typical Applications	Industrial control, robotics, and automotive network data collection

1.4 Use Case

The 10BASE-T1S-USB-ETH can be deployed in both Point-to-Point and Multidrop network topologies, connecting multiple devices to a single shared twisted-pair bus using the PLCA mechanism for collision-free transmission.

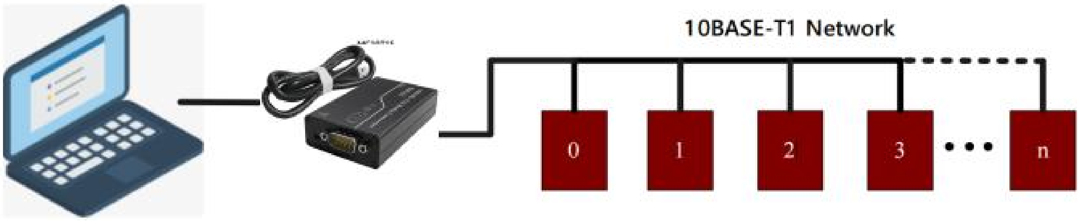
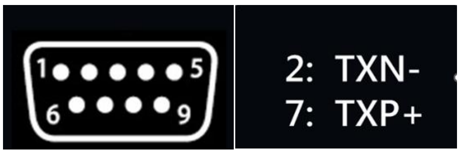


Figure 1-1. Typical use case

2. Hardware

2.1 DB9 Pin Out

The DB9 connector provides the 10BASE-T1S differential signal pair:



Ref.	Item	Description
①	DB9	**PIN 2,**TXN−,10BASE-T1S differential signal negative **PIN 7,**TXP+,10BASE-T1S differential signal positive
②	100 Ω Termination	Left position: Disable | Right position: Enable

2.2 LED Indicators

Figure 2-2. Status LED location

Status LED behavior:

LED	Behavior
③ USB LINK	Flashes during USB data communication.
④ PWR	Remains constantly on when the device is powered.

2.3 Test Case Connection

The following diagram illustrates a typical two-node test setup:

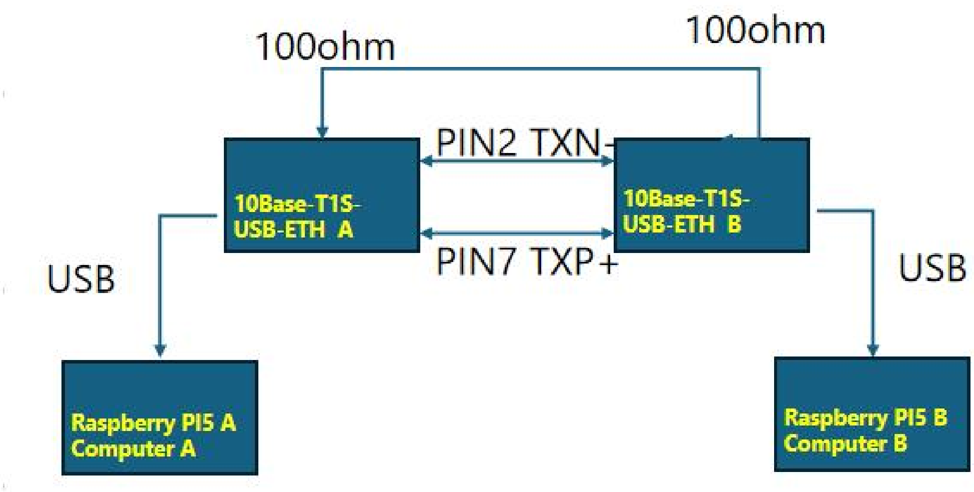


Figure 2-3. Test case connection

Note: *Enable the 100 Ω termination resistor on the terminal boards at both ends of a Point-to-Point link.*

3. Windows User Guide

Connect the hardware following Section 2.3.

3.1 Install Driver (Windows 10 or above)

Plug the 10BASE-T1S-USB-ETH into your PC. Windows will automatically recognize the device. You can find "10BASE-T1S USB Adapter" listed in Device Manager.

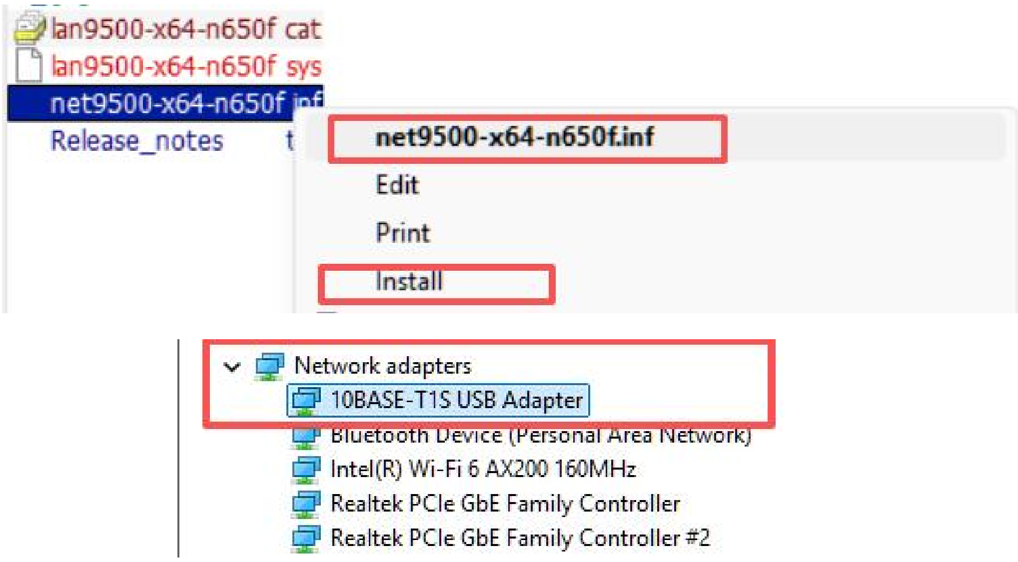


Figure 3-1. Adapter in Device Manager

3.2 10BASE-T1S USB Adapter Properties

Right-click on the "10BASE-T1S USB Adapter" entry in Device Manager and select Properties.

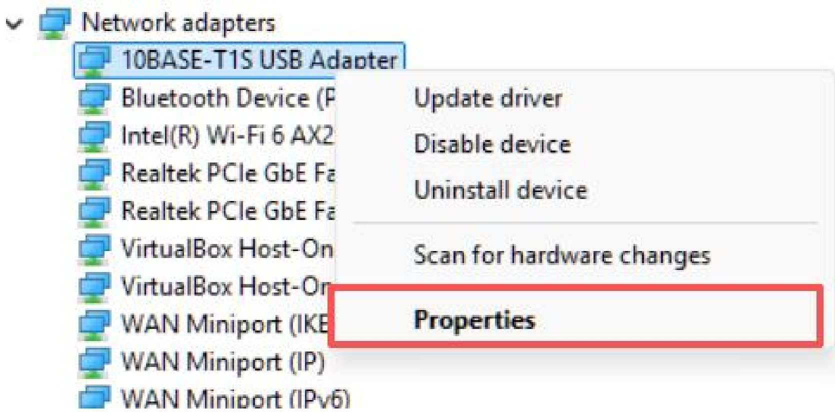


Figure 3-2. Adapter properties

3.3 10BASE-T1S PLCA Enable

In the adapter's Properties window, enable the PLCA option. PLCA (Physical Layer Collision Avoidance) must be enabled on all nodes sharing the same Multidrop bus for deterministic, collision-free operation.

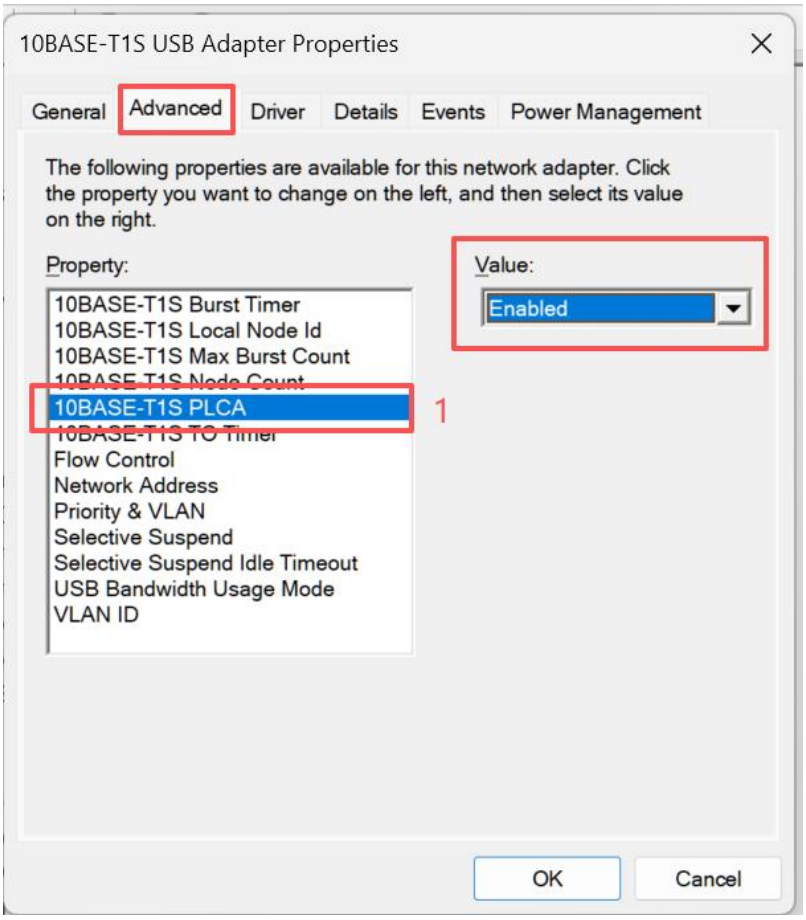


Figure 3-3. PLCA Enable setting

3.4 10BASE-T1S Local Node ID Setting

Note: *The Node ID value must be unique within the same PLCA network. Valid values start from 0, 1, 2, 3, etc.*

In this example, Computer A is configured with Node ID = 0 and Computer B with Node ID = 1.

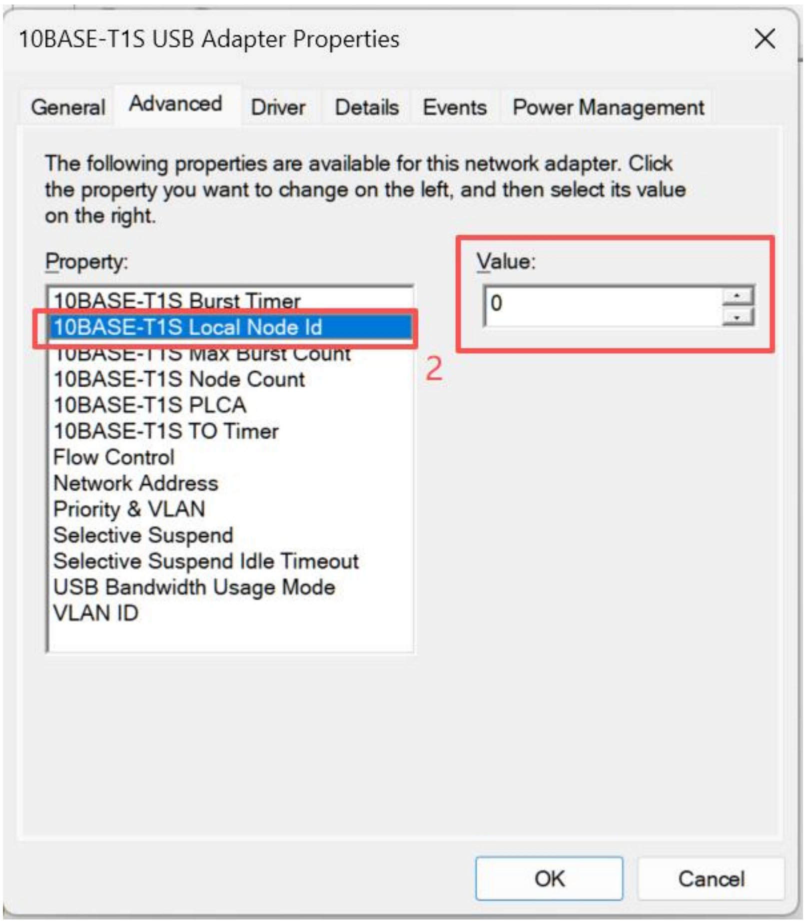


Figure 3-4. Local Node ID setting

3.5 10BASE-T1S Node Count Value

Set the network maximum Node Count. In this example, we set Node Count = 8.

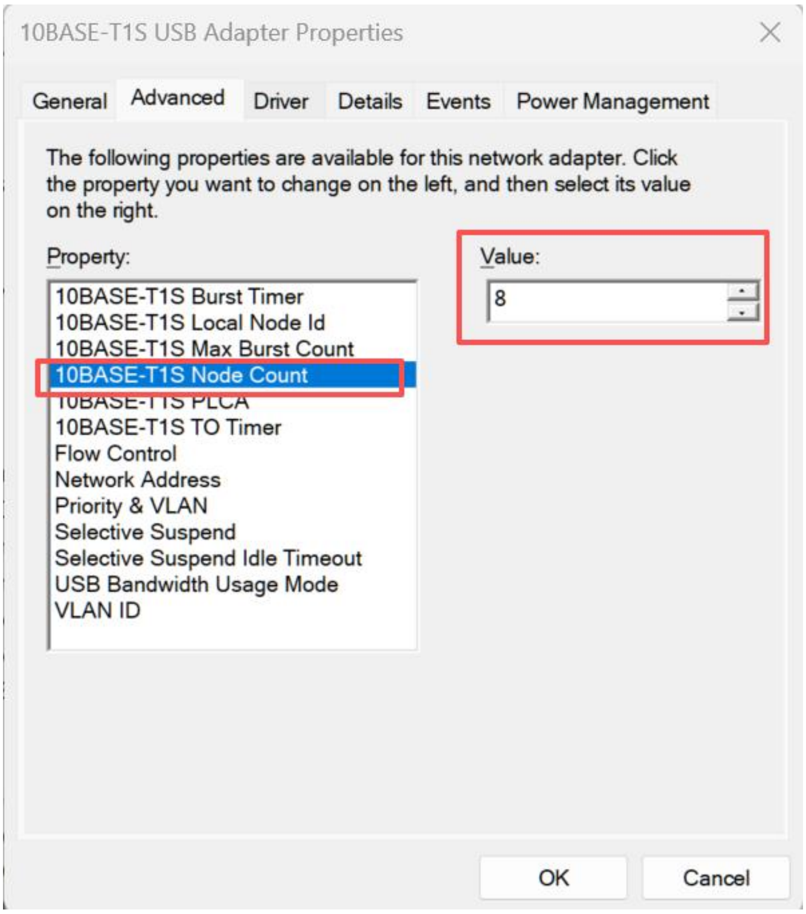


Figure 3-5. Node Count setting

3.6 10BASE-T1S USB Adapter IP Setup

Note: *Turn off the firewall on both computers before running the test.*

Locate the 10BASE-T1S USB Adapter by navigating to:

```
Control Panel \ All Control Panel Items \ Network and Sharing Center \ Change
Adapter Settings
```

Right-click the adapter and open Properties:

Then select Internet Protocol Version 4 (TCP/IPv4) and click Properties to assign a static IP:

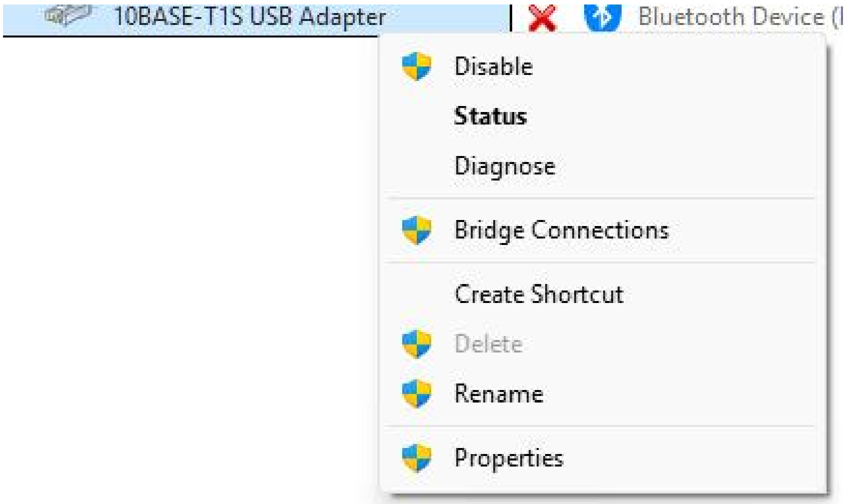
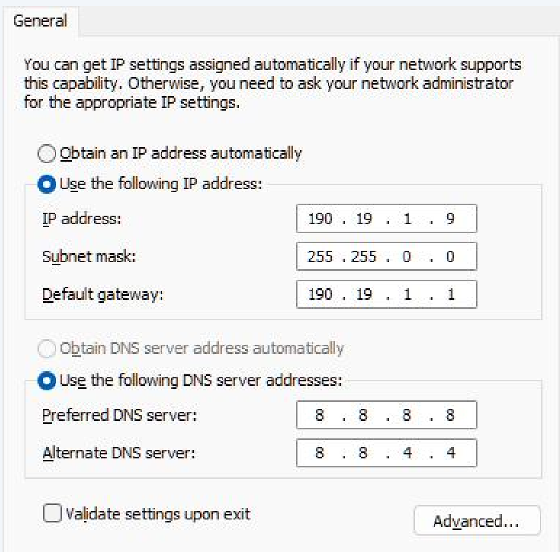
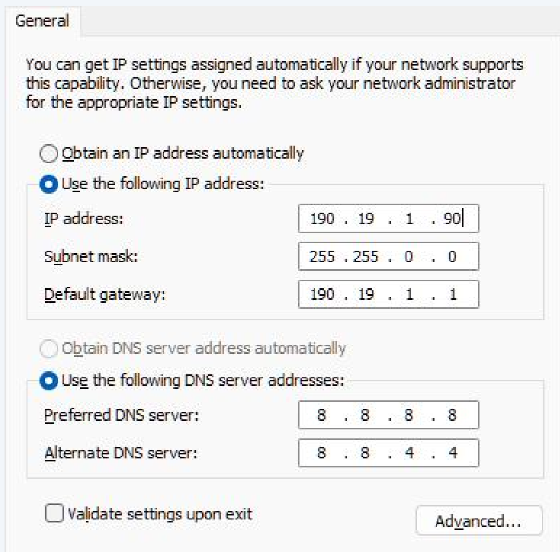


Figure 3-6. TCP/IPv4 Properties

3.7 IP Configuration





Computer A	Computer B
IP Address: 190.19.1.9 Subnet Mask: 255.255.0.0	IP Address: 190.19.1.90 Subnet Mask: 255.255.0.0

Default Gateway: 190.19.1.1 Preferred DNS: 8.8.8.8 Alternate DNS: 8.8.4.4	Default Gateway: 190.19.1.1 Preferred DNS: 8.8.8.8 Alternate DNS: 8.8.4.4

3.8 Test Report

Throughput testing uses iperf3 with the roles described below.

Scenario 1: 190.19.1.9 as Server, 190.19.1.90 as Client

On 190.19.1.9 (Server):

```
.\iperf3.exe -B 190.19.1.9 -s
```

On 190.19.1.90 (Client):

```
.\iperf3.exe -c 190.19.1.9 -B 190.19.1.90 -w 10M
```

```
[redacted] .\iperf3.exe -B 190.19.1.9 -s
-----------------------------------------------------------
Server listening on 5201
-----------------------------------------------------------
Accepted connection from 190.19.1.90, port 52251
[  5] local 190.19.1.9 port 5201 connected to 190.19.1.90 port 52252
[ ID] Interval           Transfer     Bandwidth
[  5]   0.00-1.00   sec   981 KBytes  8.03 Mbits/sec
[  5]   1.00-2.00   sec  1.08 MBytes  9.10 Mbits/sec
[  5]   2.00-3.00   sec  1.09 MBytes  9.10 Mbits/sec
[  5]   3.00-4.00   sec  1.08 MBytes  9.08 Mbits/sec
[  5]   4.00-5.00   sec  1.09 MBytes  9.10 Mbits/sec
[  5]   5.00-6.00   sec  1.08 MBytes  9.08 Mbits/sec
[  5]   6.00-7.00   sec  1.09 MBytes  9.10 Mbits/sec
[  5]   7.00-8.00   sec  1.09 MBytes  9.10 Mbits/sec
[  5]   8.00-9.00   sec  1.08 MBytes  9.08 Mbits/sec
[  5]   9.00-10.00  sec  1.08 MBytes  9.10 Mbits/sec
[  5]  10.00-10.70  sec   774 KBytes  9.09 Mbits/sec
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bandwidth
[  5]   0.00-10.70  sec  0.00 Bytes  0.00 bits/sec                  sender
[  5]   0.00-10.70  sec  11.5 MBytes  8.99 Mbits/sec                  receiver
-----------------------------------------------------------
Server listening on 5201
```

Figure 3-7. Scenario 1 test result

Scenario 2: 190.19.1.90 as Server, 190.19.1.9 as Client

On 190.19.1.90 (Server):

```
.\iperf3.exe -B 190.19.1.90 -s
```

On 190.19.1.9 (Client):

```
.\iperf3.exe -c 190.19.1.90 -B 190.19.1.9 -w 10M
```

```
PS [redacted]\iperf-3.1.3> .\iperf3.exe -c 190.19.1.90 -B 190.19.1.9 -w 10M
Connecting to host 190.19.1.90, port 5201
[  4] local 190.19.1.9 port 1049 connected to 190.19.1.90 port 5201
[ ID] Interval           Transfer     Bandwidth
[  4]   0.00-1.00   sec  11.0 MBytes  92.1 Mbits/sec
[  4]   1.00-2.00   sec  1.12 MBytes  9.44 Mbits/sec
[  4]   2.00-3.00   sec  1.00 MBytes  8.40 Mbits/sec
[  4]   3.00-4.00   sec  1.12 MBytes  9.43 Mbits/sec
[  4]   4.00-5.00   sec  1.12 MBytes  9.44 Mbits/sec
[  4]   5.00-6.00   sec  1.00 MBytes  8.38 Mbits/sec
[  4]   6.00-7.00   sec  1.12 MBytes  9.44 Mbits/sec
[  4]   7.00-8.00   sec  1.12 MBytes  9.44 Mbits/sec
[  4]   8.00-9.00   sec  1.00 MBytes  8.39 Mbits/sec
[  4]   9.00-10.00  sec  1.12 MBytes  9.44 Mbits/sec
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bandwidth
[  4]   0.00-10.00  sec  20.8 MBytes  17.4 Mbits/sec                  sender
[  4]   0.00-10.00  sec  11.4 MBytes  9.53 Mbits/sec                  receiver
```

Figure 3-8. Scenario 2 test result

4. Linux User Guide

Connect the hardware as described in Section 2.3.

Follow the installation and usage instructions provided in the BUELE GitHub repository:

```
https://github.com/buelec-tech/10Base-T1S-USB-ETH/blob/main/README.md
```

5. Packing List

No.	Item	Quantity
1	10BASE-T1S-USB-ETH Converter Unit	1 pcs
2	DB9 to Terminal Board	1 pcs